Experiment 6

AIM: Setup Nginx to handle proxy requests and load balancing

Step 1: Create Simple Node.js App

1. Create a Project

```
C:\Yash EX5>mkdir load-balance-demo

C:\Yash EX5>cd load-balance-demo

C:\Yash EX5\load-balance-demo>npm init -y
(node:12140) ExperimentalWarning: CommonJS module C:\Program
Files\nodejs\node_modules\npm\node_modules\debug\src\node.js is loading ES Module
C:\Program Files\nodejs\node_modules\npm\node_modules\supports-color\index.js using
require().
Support for loading ES Module in require() is an experimental feature and might
change at any time
(Use `node --trace-warnings ...` to show where the warning was created)
Wrote to C:\Yash EX5\load-balance-demo\package.json:

{
  "name": "load-balance-demo",
  "version": "1.0.0",
  "main": "index.js",
  "scripts": {
    "test": "echo \"Error: no test specified\" && exit 1"
  },
  "keywords": [],
  "author": "",
  "license": "ISC",
  "description": ""
}

C:\Yash EX5\load-balance-demo>npm install express
(node:7004) ExperimentalWarning: CommonJS module C:\Program
Files\nodejs\node_modules\npm\node_modules\debug\src\node.js is loading ES Module
C:\Program Files\nodejs\node_modules\npm\node_modules\supports-color\index.js using
require().
Support for loading ES Module in require() is an experimental feature and might
change at any time
(Use `node --trace-warnings ...` to show where the warning was created)
added 65 packages, and audited 66 packages in 4s
22 packages are looking for funding
  run `npm fund` for details
found 0 vulnerabilities
```

2. Create server.js

Terminal 1:

```
C:\Yash EX5\load-balance-demo>notepad index.js
```

Edit **index.js**:

```
const express = require("express");
const app = express();
const PORT = process.env.PORT || 3000;
```

```
app.get("/", (req, res) => {
  res.send(`Response from Server running on port ${PORT}`);
});
app.listen(PORT, () => {
  console.log(`Server running on port ${PORT}`);
});
```

3. Run the application node server.js

```
C:\Yash EX5\load-balance-demo>node index.js
Server running on port 3000

C:\Yash EX5\load-balance-demo>node index.js
Server running on port 3000
```

4. Create a Folder Load balancing

Terminal 2:

```
C:\Yash EX5\load-balance-demo>notepad index1.js
```

Edit **index1.js**:

```
const express = require("express");
const app = express();
const PORT = process.env.PORT || 3001;

app.get("/", (req, res) => {
  res.send(`Response from Server running on port ${PORT}`);
});
app.listen(PORT, () => {
  console.log(`Server running on port ${PORT}`);
});
```

Step 2: Run Multiple Servers

Open 2 terminals:

Terminal 1:

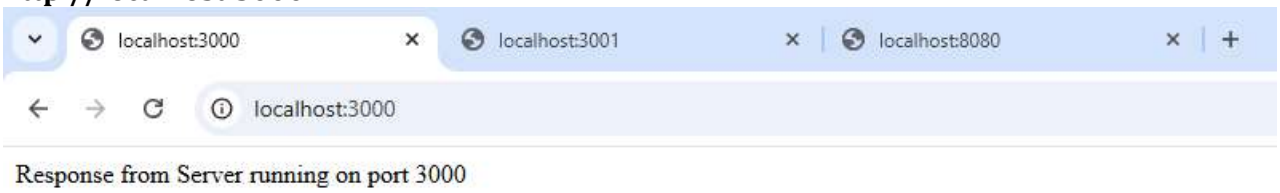
```
C:\Yash EX5\load-balance-demo>node index.js
Server running on port 3000

C:\Yash EX5\load-balance-demo>node index.js
Server running on port 3000
```

Terminal 2:

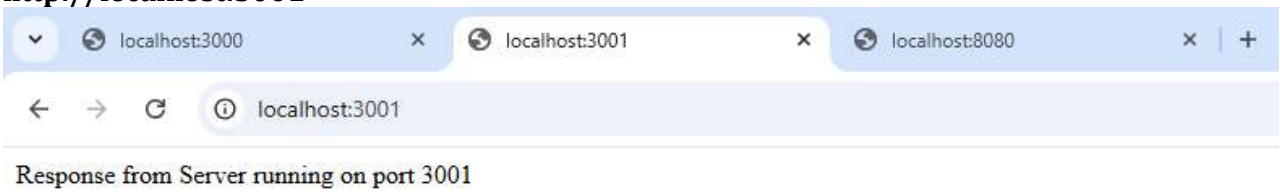
```
C:\Yash EX5\load-balance-demo>node index1.js
Server running on port 3001

C:\Yash EX5\load-balance-demo>node index1.js
Server running on port 3001
```

So we have now:

http://localhost:3000



http://localhost:3001



Step 3. Configure NGINX for Load Balancing

(Open nginx.conf and Modify http block)

```
events {
    worker_connections 1024;
}

http {
upstream myapp {
     server localhost:3000;
     server localhost:3001;
     }

    server {
        listen 8080;
        server_name localhost;

        location / {
            proxy_pass http://myapp;
        }
    }
}
```

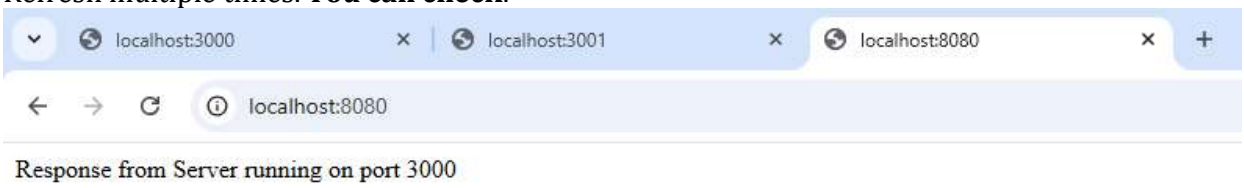
Step 4: Restart NGINX

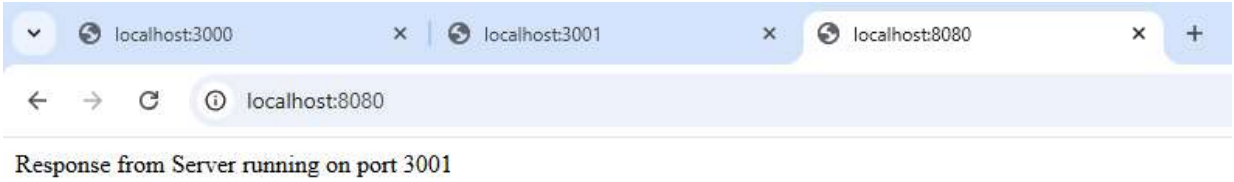
Terminal 3:

```
C:\nginx>nginx -s stop
C:\nginx>nginx
```

Now open: **http://localhost:8080**

Refresh multiple times. **You can check**:





Step 5: Make Server 1 Slow

Modify Server running on PORT 3000:

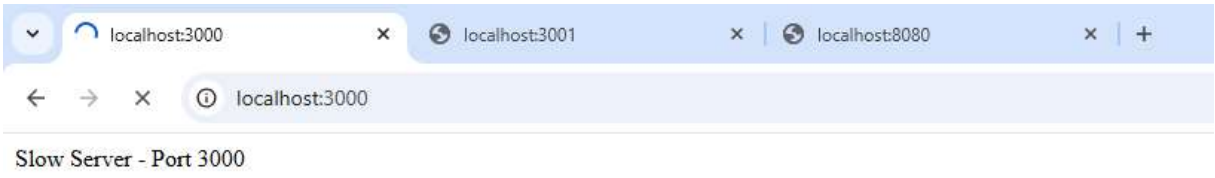
```
const express = require("express");
const app = express();

const PORT = process.env.PORT || 3000;
app.get("/", (req, res) => {
setTimeout(() => {
res.send("Slow Server - Port 3000");
}, 5000);
});

app.listen(PORT, () => {
 console.log(`Server running on port ${PORT}`);
});
```

Keep Server 2 normal:

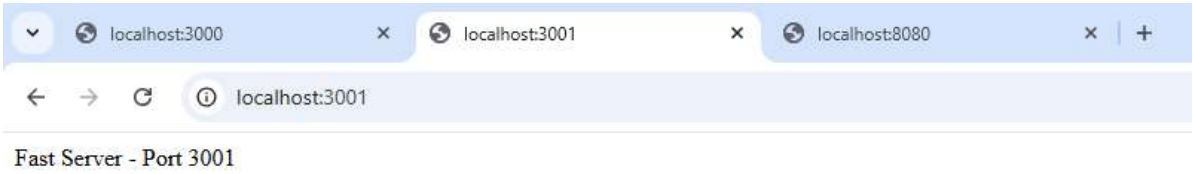
```
const express = require("express");
const app = express();

const PORT = process.env.PORT || 3001;
app.get("/", (req, res) => {
res.send("Fast Server - Port 3001");
});

app.listen(PORT, () => {
 console.log(`Server running on port ${PORT}`);
});
```

Step 6. Do Testing

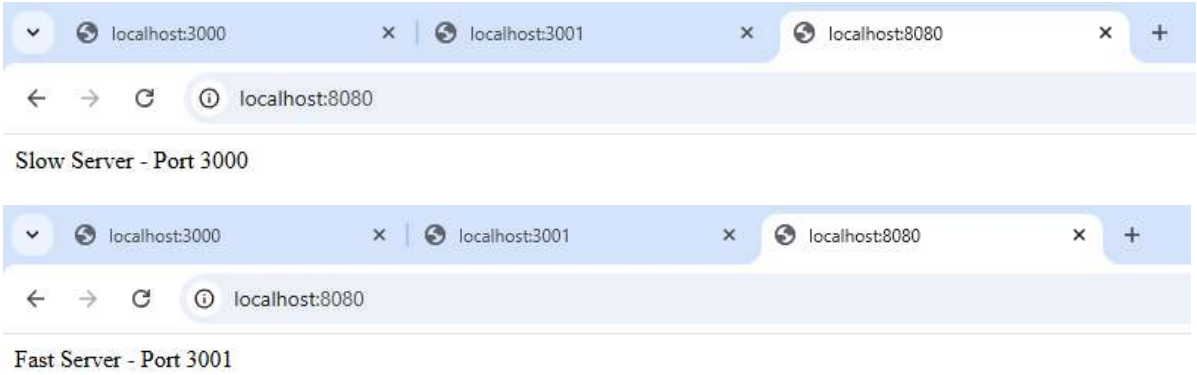
Server 1 need 5 second to load on 3000



Server 2 as it is Fast on 3000



Open: **http://localhost:8080** Now refresh quickly multiple times.
Even though server 3000 is slow, NGINX still keeps sending every alternate request to it.
Server 1 need 5 second to load
Server 2 as it is Fast



Step 7: Now Change default algo of nginx(i.e. from round robin) to Least Connections and test again

Modify nginx.conf file:

```
events {
    worker_connections 1024;
}
http {
upstream myapp {
     least_conn;
     server localhost:3000;
     server localhost:3001;
     }
    server {
        listen 8080;
        server_name localhost;
        location / {
            proxy_pass http://myapp;
        }
    }
}
```

Restart NGINX:

```
C:\nginx>nginx -s stop

C:\nginx>nginx
```

Open: http://localhost:8080 Refresh quickly multiple times quickly.
When slow server (3000) is busy for 5 seconds:
It has an active connection.
So NGINX sees:
Server 3000 → 1 active connection
Server 3001 → 0 active connections

So it sends new requests to:
Fast Server (3001)